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Original Article

Effects of Maize Lethal Necrosis (MLN) Disease on Maize Output and Food Security Among Smallholder Farmers in Bomet County, Kenya

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ABSTRACT

Maize is one of the leading cereals produced and consumed in Kenya and is very vital to food security. In the recent past, maize production has encountered many problems including poor input use, drought, pests, and diseases. Maize production in Kenya has been particularly devastated by Maize Lethal Necrosis (MLN), a viral disease which unexpectedly infested the maize crops in 2011. The disease has caused major economic and social losses to maize farmers and has spread to other neighbouring countries. Data on 3368 households was collected using a two-stage stratified cluster sampling method. The estimation procedure involves the use of propensity score matching to determine the effect of MLN on household food security and maize output. The results of this study indicate that MLN disease reduces household food security by about 16.6 percent, while it reduces maize output by about 10 (90 kgs.) bags per acre. The study therefore recommends that the government and stakeholders design appropriate policies for restoring plant health which include maize-resistant varieties, training of smallholder farmers on good agronomic practices and growing of alternative crops in order to break the disease cycle thus ensuring food security.

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INTRODUCTION

Agriculture is the engine of Kenya's economic growth and the most important source of rural livelihood. It contributes directly about 26 percent and indirectly another 25 percent to the Gross Domestic Product of Kenya's economy (GoK, 2010). It also contributes about 65 percent to the total annual export volume and employs more than three-quarters of the Kenyan labour force (GoK, 2010). Kenya's agricultural sector is predominantly farmed by smallholders who are responsible for producing over 70 percent of maize output (GoK, 2010). Agriculture is important in food security and poverty reduction and an increase in agricultural production can increase rural income and lower the poverty of poor smallholder farmers as well as grow the Kenyan economy. Kenya's Vision 2030 underscored the importance of transforming the agriculture sector from a subsistence to a commercially oriented sector to boost economic growth, and food security and expand agricultural exports (D'Alessandro et al., 2015).

Maize is a very important food security crop in the world and provides food to 4.5 billion people living in about 100 developing countries. In Africa, maize contributes 20 percent of food calories (Shiferaw et al., 2011). Africa uses 95 percent of the maize produced as food thus making maize a very important food security crop. In Sub-Saharan Africa, maize is produced in over 25 million hectares mostly by small-scale farmers who harvest about 38 million tons of grains annually. Maize production in Sub-Saharan Africa represents 34 percent of all cereals produced. About 85 percent of maize produced in Eastern and Southern Africa is consumed by people (Fleet & Mashingaidze, 2016).

Maize grows well in most agroecology zones in Kenya and is grown on over half of the cultivated land by 98 percent of 3.5 million smallholder farmers under rain-fed conditions (Kirimi, 2012). Maize accounts for 28 percent of the total value of Kenyan agricultural production and 14 percent of smallholder income (Nyoro et al., 2005). The average yield of maize is 1.5 tons/ha to 3 tons/ha (GoK, 2010). On average smallholder farmers harvest up to 16 bags while large-scale farmers harvest 20 bags (Kirimi et al., 2018). In 2011, about 2 million hectares of land were cultivated and planted with maize in Kenya which yielded approximately 38 million bags of dry maize and about 5 million bags of green maize, all with a value of about 87.8 billion Kenyan Shillings (KARI, 2012).

Maize constitutes a major staple food crop, contributes to food security and satisfies the nutritional needs of many people by providing about a third of calorie requirements (Kirimi, 2012). In Kenya, maize consumption per person is 103 Kg per annum (De Groote et al., 2002) and per capita maize consumption is 171 g/person/day which is the highest in East Africa (Ranum et al., 2014). Consequently, food security in Kenyan households is always measured on the basis of the availability of maize to meet food demand (Midega, 2013). Rural households depend on maize production for commercial purposes and for food while the growing urban population who are maize net buyers depend on maize for food at tolerable prices (Nyoro & Kirimi, 2004).

Low maize production has been attributed to climate change and further worsened by biotic and abiotic factors. Maize cultivation in Kenya has been declining due to maize lethal necrosis (MLN), a viral disease which occurred suddenly in South Rift Kenya in 2011 and has since spread to other maize-growing regions in the country (Wangai et al., 2012). MLN originated in the United States of America (USA) with the first case reported in

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Kansas in 1978 (Jardine, 1978) and spread to China in 2009 (Wang et al., 2017). The MLN is caused by interaction between maize chlorotic mottle virus and either wheat streak mosaic virus or sugar cane mosaic virus. In Kenya maize lethal necrosis occurs mostly because of the interaction between maize chlorotic mottle virus and sugar mosaic virus. Infected plants develop yellow spots on the leaves. Infection in the later stage of the growth leads to mottling on the leaves which eventually dries up. Infected maize plants produce small cobs with or without grains (Boa, 2014).

Mahuku et al. (2015) estimated maize loss valued at 52 US dollars in 2012 or 126000 tons which represent 90 percent of total maize production lost. In 2013 approximately 26000 ha of maize valued at about 2 billion Shillings was lost to the disease and represents a reduction of more than 1 million tons of maize (Virginia, 2015). The nominal value of loss due to MLN was $187 million, most of which was concentrated in Western Kenya and Rift Valley regions where in Western Kenya more than half of the farms were affected (De Groote 2015). Marenya et al. (2018) estimated maize loss to be $198 million. The disease has caused tremendous suffering to over 90 percent of people who depend on maize for food and income in Kenya. During the 2014/2015 season about 60,000 hectares of maize were affected, representing a 10 percent drop in maize production further complicating the food security status of the people (Mwatuni et al. 2017).

The interruption of maize production poses a serious food insecurity problem to millions of households depending on it (Gitonga, 2014). The government of Kenya and researchers have yet to evaluate and quantify the effect of this disease on food security and maize output. Estimating the impact of MLN on food security would help to design appropriate policies for restoring plant health and food security of smallholder farmers (Isabirye & Rwomushana, 2016). The objective of this study is to determine the effect of maize lethal necrosis on maize output and food security of smallholder farmers. Therefore, this study will help shed light on the effect of MLN on food and nutrition security and enable government agencies to undertake measures to assist farmers in controlling the menace of this disease.

RESEARCH MATERIALS AND METHODS

The Survey Data and Variables

The study was conducted in Bomet County, Kenya. The county has a total area of 2037.4 Km^2 and is made up of 5 Sub-Counties which include; Konoin, Bomet East, Bomet Central, Sotik and Chepalungu. According to the 2019 population and housing census, Bomet County had a population of 875,689 people (KNBS, 2019). Bomet County was purposively chosen because agriculture, specifically maize farming is the mainstay of the County and in addition, the County was most affected by the MLN disease (De Groote et al.,2015). This study used secondary data which was collected in 2014 using 2 stage stratified cluster sampling method. In the first stage, 350 rural clusters were picked and in the second stage, 20 households were randomly selected from the clusters using a systematic random sampling method. In total 3,368 households were sampled. The data covered the 2013/2014 cropping season. The survey data is rich with information on household socio-economic and production characteristics of the smallholder farmers.

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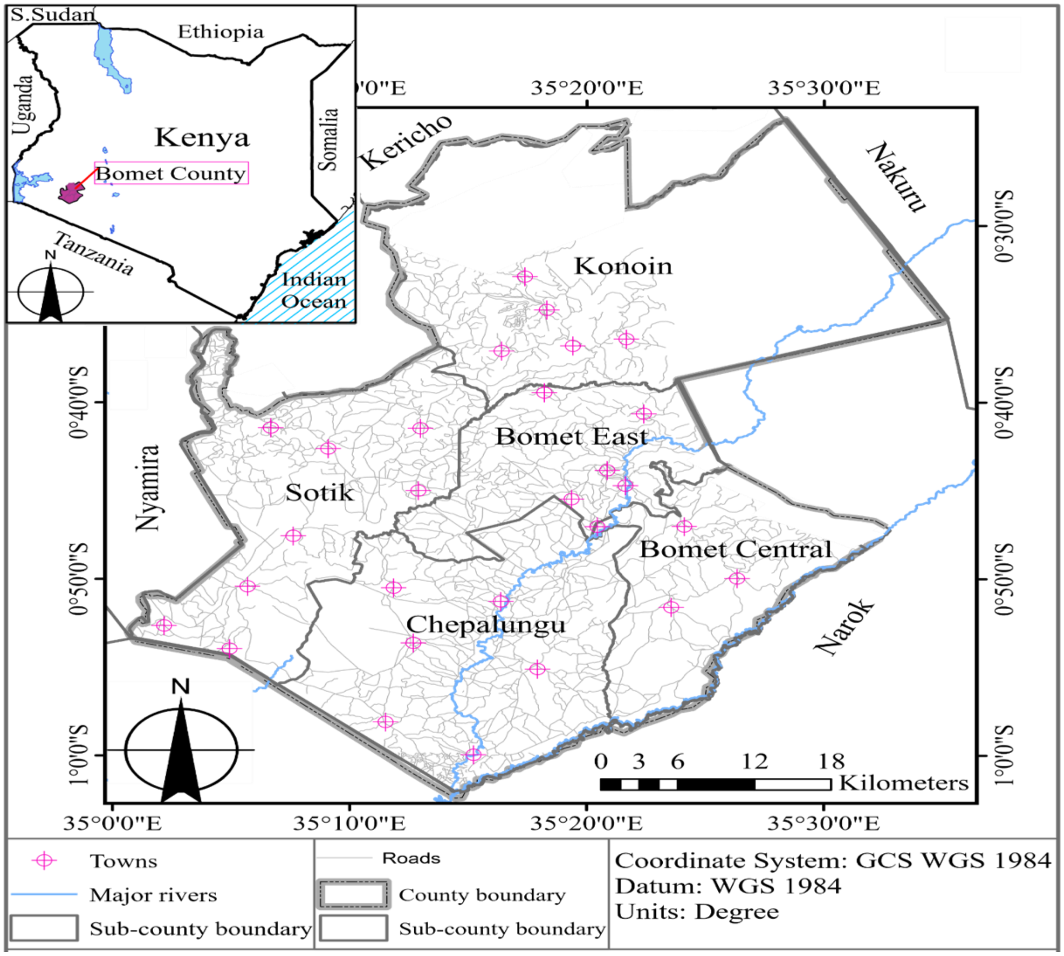
Figure 1: Map of Kenya indicating the study areas.



Food Security

Food security is a global concern where many people worldwide lack enough food to eat affecting their ability to carry out day-to-day activities. There are five methods of measuring food insecurity. Four of the methods are indirect methods and include; the FAO method, household expenditure survey, dietary anthropometry and intake assessment. The sole direct method for food security is one built on an experienced-based food insecurity scale. All methods are complementary to each other and each method is applied based on the question asked (Pérez-Escamilla & Segall-Corrêa, 2008).

In this paper household food security survey module, six items were used. This method of measuring food insecurity was developed by researchers at the National Center for Health Statistics in the United States and was found to have a 97.7 percent level of accuracy in predicting food insecurity (Blumberg et al.,1999). Responses "often" or "sometimes" and "yes" are coded as positive. The summation of the six questions forms categories of food security into which households can fall, 0-1 represents high food-secure household households,2-4 represents low food-secure households and 5-6 represents very low-secure households (Blumberg et al., 1999). In this study, low food security and very low food security were merged to form food insecurity to have binary food security measurement, food secure and food insecure.

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Empirical Estimation

The effect of the MLN on food security and maize output was estimated using a propensity score matching approach. A propensity score is "the conditional probability of assignment of a particular treatment given a vector of observed covariates" (Rosenbaum & Rubin, 1984). In PSM participants are matched with non-participants. In this case, maize farmers who were affected by MLN are matched with farmers who produced maize but were not affected by the MLN disease. For this method to be effective, unobserved factors should not affect participation and there should be enough common support on propensity scores across both participants and non-participants (Khandker et al., 2009).

Outcomes of participants and non-participant farmers with the same propensity scores are compared. Propensity score matching builds a statistical group for comparison that is based on the model of the probability of participating in the treatment T based on observed characteristics X.

$$P(X) = Pr(T = 1) \ldots\ldots (1)$$

According to Rosenbaum & Rubin (1984) matching $P(X)$ is as good as matching on X under certain given assumptions. In this study dependent variable of interest is binary in nature. Value 1 is taken by farmers who grew maize and was affected by MLN while value 0 is taken by farmers who grew maize and were not affected by the MLN disease. In assessing an intervention, it is prudent to infer what would have happened if the farmers were not affected by the MLN disease. Evaluation of program impact require determination of average treatment on the treated (ATT) which is simply difference between the treated farmers and counterfactual. According to Khandker et al. (2009), "Counterfactual outcomes for participants had they not been exposed to the program".

Given the binary nature of the dependent variable, the treatment indicator T_i Equal one if smallholder farmer was affected by MLN and zero otherwise. The potential outcome can be indicated as $Yi(Di)$ for each individual farmer i, $i = 1,2,3, \ldots, n$. The treatment effect for individual farmers will be written as follows:

$$T_i = Yi(1) - Yi(0) \ldots\ldots (2)$$

However, it is not possible to estimate T_i, hence average treatment effect on treated is developed and specified as

$$T_{ATT} = E[T|D = 1 = E[Y1|D = 1] - E[Y(0)|D = 1] \ldots\ldots (3)$$

The counterfactual average of the farmers affected by MLN is

$$-E[Y(0)|D = 1]$$

$$ATT = E[Yi(1)|D = 1] - E[Yi(0)|D = 0] = T_{ATT} + E[Y0|D = 1] - E[Y(0)|D = 0] \ldots\ldots (4)$$

Where ATT is referred to as self-selection bias and the true parameter of T_{ATT} is

$$E[Y(0)|D = 1] - E[Y(0)|D = 0] = 0 \ldots\ldots (5)$$

$$ATT = E[Y(1) - Y(0)] \ldots\ldots (6)$$

Common support is indicated as

$$(Overlap\ ) \ 0 < p(D = 1|X) < 1 \ldots\ldots (7)$$

Therefore, ATT is just the difference in means of outcomes over common support, appropriately weighted by the propensity score distribution of participants. The PSM model is specified as

$$T_{ATT} = Ep(x)|D = 1\{E[Y(1)|D = 1, p(x)] - E[Y(0)|D = 0, p(x)]\} \ldots\ldots (8)$$

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To estimate the propensity score for each household in the sample, we use a logistic regression in which the propensity score is a function of X as follows.

$$P_r(D_i = 1|X) = Q(X\beta) = Q(\beta_0 + \beta_1 x_1 + \cdots \beta_k x_k), \quad (9)$$

where x_i is the variables in vector X and β_i are the corresponding coefficients while Q represents the logistic regression function. The cumulative distribution function for this logistic regression is given by:

$$\Lambda(X\beta) = \frac{\exp(X\beta)}{1+\exp(X\beta)} \quad (10)$$

Upon estimation, the predicted probabilities should be between 0 and 1. The predicted probability for each observation in the sample is its propensity score. To establish a "common support," the sampled households are excluded in the control counties for which the propensity score is zero and lower or higher than any of those in the treatment sample. For each household in the treatment sample, we find the household in the retained control sample that has the closest propensity score, as measured by the absolute difference in the scores. Based on the estimated equally spread-out interval of propensity scores, treated and control households are divided into sub-samples, testing each interval to ensure the mean scores of treated and control are the same. This process continues until the interval of propensity score of the treated and controlled are the same which is the necessary condition for balancing (Becker & Ichino, 2002).

Because of the low likelihood of finding pairs from observations in the treatment and control groups with exactly the same propensity score, several algorithms have been proposed and used, which include the nearest neighbour matching, calibre or radius method, stratification and kernel method. For each observation in the treatment group, the nearest neighbour method searches for the observations in the control group with nearby propensity scores. Using the five nearest neighbours is a common practice. Upon forming the block of nearest neighbours, the mean value of each outcome indicator is calculated. The difference between that mean and the actual value for the treated observation is the estimate of the benefit of the observation that is attributable to the program. The mean of the individual benefits is the average treatment effect on the treated.

Applying the PSM method to the survey data, the impact of MLN is evaluated on two outcome indicators: households' food security status and maize output. The survey data have enough information to determine whether the treated and control households are food secure, or food insecure. PSM model addresses problems of hidden bias which could otherwise lead to biased estimates. This approach has been used by several scholars including, Wabwile et al. (2016) to determine the effect of improved sweet potato varieties on food security, Abebaw, Fentie, & Kassa (2010) to analyse the impact of a food security program on household food consumption in North-western Ethiopia and Tirivayi & Groot(2017) to determine the impact of food assistance on dietary diversity and food consumption among people living with HIV/AIDS.

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Table 1: **Description of the variables to be used in the model**

Variable	Description and measurement
Food security	D=1 if the household head is food secure, 0 otherwise
Maize Lethal Necrosis	D=1 if HH was affected by MLN, 0 otherwise
Age	Age of the household head in years
Male	D=1 if HH head is male,0 otherwise
Primary Education	D=1 if HH head if household head completed primary education
Secondary Education	D=1 if HH head if household head completed secondary education
College Education	D=1 if HH head if household head completed college education
Household size	Number of household members in the household
Land size	Number of acreages of the household land
Group membership	D=1 if the HH head belongs to an agricultural group, 0 otherwise
Saving Account	D=1 if HH have a savings account, 0 otherwise
Off-farm income	Amount earned in Kenya shillings
Distance to market	Distance in kilometres to the market
Land tenure	D=1 if the household has a title, 0 otherwise
Farm experience	Number of years households have been farming.

Note**:** HH stand for Household, MLN stand for Maize Lethal Necrosis

RESULTS AND DISCUSSIONS

The mean age for household heads affected by MLN is about 50 years while those not affected by MLN are also about 50 years. The results of the two-tailed t-test show that the age of the household head was not statistically significant showing that the ages between those affected by MLN and those who were not affected were equal. The mean number of households of those affected by MLN is about 6 members while non-MLN households have a mean number of 5 members. The two-tailed t-test indicates that household size was statistically significant at 5 percent and therefore we conclude that the household size of those affected by MLN was bigger than households not affected by the MLN. The mean value of assets of households affected by MLN is Ksh 199,170 while those not affected by MLN is Ksh 184,110. The results from the two-tailed t-test show that asset value was not statistically significant meaning that those affected by MLN and those who were not affected by the disease had equal asset value. The mean asset value between the two groups is not different. The mean distance to the market for those affected by MLN is 6 kilometers while those who were not is 4 kilometers. The results from the two-tailed t-test that the distance to the market was statistically significant at 1 percent and therefore we conclude that the difference in distance to the market of those affected by MLN and those who were different from zero.

The mean land size of households affected by MLN is 4 acres while those not affected by MLN is 7 acres. The results from the tailed t-test show that the land size was not statistically significant meaning that those affected by MLN and those who were not affected by the disease had no different land size. The mean off-farm income of households affected by MLN is Ksh 125,325 while those not affected by MLN is Ksh 140505. The results from the tailed t-test show that the off-farm income was not statistically significant meaning that those affected by MLN and those who were not affected by the disease had a similar amount of off-farm income. The mean farmer experience between households affected by MLN is around 23 years while for those not affected by MLN is also about 23 years. The results from the tailed t-test show that the farmer experience was not statistically significant meaning that those affected by MLN and those who were not affected by the disease had similar amounts of farming experience.

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Table 2: Summary of continuous socio-economic characteristics of households

Variable		Mean				
	Observation	**MLN**	**No MLN**	**Overall**	**t-stat**	**p-Value**
Age	3,366	50.14	49.87	49.896	-0.276	0.7826
Household size	3,368	5.86	5.49	5.524	-2.286**	0.0223
Asset Value	3,366	199170.6	184110.1	185362.9	-0.5108	0.6096
Land size	3,368	4.25	7.42	7.157	0.362	0.7171
Distance to market	3,368	6.08	4.46	4.595	-4.86***	0.0000
Off-farm income	2,106	125325	140505.9	139215.6	0.440	0.6597
Farmer experience	3,180	22.98	22.81	22.82	-0.168	0.8666

Note: *, ** and *** stand for significance level at 10%,5% and 1% respectively

Impact of Maize Lethal Necrosis

The main objective is to test the specific hypothesis – the effect of MLN is negative and significant for all two outcome variables; food security and maize output. A logistic regression model was used to estimate propensity score matching and this assisted in matching the algorithm between those households affected by MLN disease and those not affected. Table 4 provides a description of the variables and overall summary statistics.

Table 3: **Logit model estimation results for propensity score matching.**

Variables	**MLN and Non-MLN**	**Robust standard errors**
Age	0.0006	0.0046
Gender	0.2282**	0.1127
Primary Education	-0.3350***	0.1255
Secondary Education	-0.4973***	0.1556
College Education	0.1134	0.1945
Household size	0.0048	0.0158
Land size	-0.0001	0.0006
Group Membership	0.0748	0.0832
Saving account	0.0642	0.0838
Log of farm income	0.0017	0.0297
Distance to market	0.0233***	0.0054
Land tenure	0.0157	0.0305
Extension visit	-0.0797	0.1008
Farm experience	0.0004	0.0045
Constant	-1.5384	0.4006***
No. of Observations	2221	
Wald Chi-Sq. Stat.	47.19	
Prob > Chi-sq.	0.0000	

Note: *, ** and *** stand for significance level at 10%,5% and 1% respectively

The estimated results (Table 4) using PSM indicate that MLN has a significant and negative effect on food security and maize output (90kg bags per acre) on all the propensity score estimators. The results show that MLN disease reduced food security by 16.6% based on the Nearest Neighbor Estimator (NNM) and 17.4 percent based on the Kernel Matching Method (KMM) compared to counterparts that are maize producers not affected by MLN. The findings also indicate that MLN disease reduced maize output by 10 (90 Kg) bags based on KMM and 10 (90 Kg) bags based on NNM.

These results depict how the disease affects the food security of smallholder farmers, especially noting that the household food security status of most Kenyans is based on the availability of maize crops.

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People need nutritious and healthy food supply to move out of poverty, however, the food security of the vulnerable populations in the world is threatened by plant disease outbreaks (Ristaino et al., 2021). These findings corroborate those of Rayfuse & Weisfelt (2012) who found out that pathogens attacking crops are responsible for an 11 percent reduction in the global production of maize. In East Africa, cassava mosaic virus reduced cassava output by 80 percent to 90 percent, striga weed affected about 5 million hectares of cereals while in Uganda banana wilt threatened the food security of about 70 million people when it reduced banana production by 50 percent at the beginning of 21 century (Vurro et al., 2010). According to Rizzo et al. (2021), plant diseases and pests affect the safety and availability of plants to both livestock and humans. Plant diseases and pests can lead to up to 30 percent loss in world-important staple crop production and billions of dollars in the loss in food production.

In the last 40 years, food production doubled due to pest and disease management, however, between 10 percent to 16 percent of the total harvest is still lost to pathogens (Chakraborty & Newton, 2011). This result also corroborates those of Allen (2006) who found out that worldwide every year, about 16 percent of crops are lost to pests and diseases which affect food production hence leading to food insecurity. GoK (2010) found out that crop diseases sometimes reduce crop production by 50 percent or total crop failures. According to De Groote et al.(2015), MLN in Kenya was responsible for the loss of 0.5 million tons of maize per year representing 22 percent of total production loss with the disease affecting Western Kenya the most leading to a loss of 58 percent of all maize. Maize Lethal Necrosis was expected to reduce maize production in Kenya by 10 percent during the 2014/2015 cropping season (Gitonga, 2014). According to Mahuku et al. (2015), MLN disease reduced maize grains by up to 90 percent during the 2012 cropping season which translated to 126,000 metric tons valued at USD 52 million. In Eastern Uganda, MLN caused an average yield loss of 50.5 percent valued at $332 per hectare (Kagoda et al., 2016).

Table 4: **Average treatment effects on the treated (ATT) - results from the Propensity Score Matching (PSM) method.**

Matching Algorithm	MLN.	No MLN.	ATT	t-values
ATT Food Security				
NNM	340	1693	-0.166*** (0.040)	-4.115
KMM	194	1995	-0.174 *** (0.036)	-4.874
ATT Maize Output (90 Kg bag Per Acre)				
NNM	340	1672	-10.109*** (4.451)	-2.271
KMM	194	1995	-10.382 *** (2.431)	-4.271

Note: *, ** and *** stand for significance levels at 10%,5% and 1% respectively. ATT – Average impact of Treatment on Treated. Figures in parentheses are standard errors. NNM = Nearest Neighbor Matching and KMM = Kernel Matching Method.

CONCLUSION AND POLICY RECOMMENDATION

This paper examines the effect of maize lethal necrosis on food security and maize output of smallholder farmers in Kenya, and it specifically determines the effect of maize lethal necrosis on food security and maize output among small farmers. This study finds that maize lethal necrosis reduces food security status and maize output of the households by 16.6 percent and 10 (90 kg) bags based on NNM. Given that maize is the leading food security crop in Kenya, more maize varieties which are resistant to disease need to be encouraged. The government also need to educate smallholder farmers on good agronomic practices and the growing of alternative crops in order to break the disease cycle and also to avoid farmers running into losses every year in input purchases. The government need to intensify research in order to eradicate the disease once and for all.

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